



**SAPTHAGIRI NPS**  
**UNIVERSITY**  
UNMATCHED EXCELLENCE, UNLIMITED POTENTIAL

## **SCHOOL OF APPLIED SCIENCE**

A Case Study Report

On

“ Agile Methodology”

Submitted partial fulfillment of requirement for the II semester  
Course & Code: Agile Methodology, 25MCAGL202

**Master of Computer Applications (MCA)**  
Specialization in AIML

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# 1. What is socio-technical system?

A **Socio-Technical System (STS)** is a system that consists of both **technical components** (hardware, software, machines, networks) and **social components** (people, organizations, policies, procedures, and culture) working together to achieve a common objective.

## Formula

**Socio-Technical System = Technical System + Social System**

## Components of a Socio-Technical System

### 1. Technical Components

These include all technological resources.

Examples:

- Software applications
- Hardware
- Databases
- Networks
- Cloud infrastructure
- Security systems

### 2. Social Components

These involve human and organizational factors.

Examples:

- Users
- Customers
- Developers
- Managers
- Business rules
- Organizational culture
- Government regulations

## Characteristics

- Involves people and technology.
- Components are interdependent.
- Changes in technology affect people.
- Human decisions influence system performance.
- Requires collaboration between technical and non-technical stakeholders.

## Examples

### Banking System

Technical Part:

- ATM software
- Banking servers
- Mobile banking app
- Database

Social Part:

- Customers
- Bank employees
- Government regulations
- Security policies

### **Socio-Technical Systems in Agile**

Agile emphasizes:

- Customer collaboration
- Continuous feedback
- Team communication
- Frequent delivery
- Adaptability

Instead of focusing only on software, Agile also improves how people interact with the system.

### **Advantages**

- Better user satisfaction
- Improved communication
- Faster problem-solving
- Easier adoption of new systems
- Higher productivity

### **Disadvantages**

- Difficult to manage human behaviour
- Higher implementation cost
- Requires continuous communication

## **2. Explain the difference between complex system and system Engineering?**

A **Complex System** consists of many interconnected components whose interactions make the system difficult to predict and manage.

Examples:

- Internet
- Air traffic control
- Smart city
- Railway network

Characteristics:

- Large number of components
- Dynamic behaviour
- Difficult to predict
- Constantly changing

### **What is System Engineering?**

**System Engineering** is a structured engineering process used to design, develop, test, integrate, deploy, and maintain complex systems throughout their life cycle.

It ensures that all parts of a system work together efficiently.

**Difference Table****Complex System**

Refers to the system itself

Naturally complicated

Difficult to predict

Has many interacting components

Focuses on system behaviour

Example: Internet

**System Engineering**

Refers to the process of building the system

Systematic engineering approach

Aims to reduce complexity

Coordinates development activities

Focuses on design and management

Example: Developing an airline reservation system

**Relationship**

System Engineering is used to build and manage Complex Systems.

Example:

- Requirement analysis
- System design
- Hardware integration
- Software development
- Testing
- Maintenance

**3. Explain system procurement, system development, system operation?**

A software system passes through three major stages:

1. System Procurement
2. System Development
3. System Operation

**A. System Procurement**

System Procurement is the process of purchasing software, hardware, cloud services, or complete systems from external vendors instead of developing them internally.

**Activities**

- Identify business needs
- Prepare requirements
- Request proposals (RFP)
- Evaluate vendors
- Select supplier
- Negotiate contract
- Purchase system

**Agile Procurement**

Traditional procurement:

- Fixed requirements
- Fixed budget
- Long contracts

Agile procurement:

- Flexible contracts
- Incremental delivery
- Customer involvement
- Frequent reviews
- Continuous improvements

Example:

Instead of buying an entire ERP system at once, the vendor delivers modules sprint by sprint.

### **Advantages**

- Saves development time
- Expert solutions
- Lower development effort
- Faster deployment

### **Disadvantages**

- Vendor dependency
- Licensing costs
- Limited customization

## **B. System Development**

### **Definition**

System Development is the process of designing, building, testing, and deploying a software or hardware system.

It transforms requirements into a working product.

### **Agile Development Phases**

#### **1. Requirement Gathering**

Understand customer needs.

#### **2. Planning**

Select user stories for Sprint.

Output:

Sprint Backlog

#### **3. Design**

Create architecture and UI.

#### **4. Development**

Develop features during Sprint.

#### **5. Testing**

Perform:

- Unit Testing
- Integration Testing
- System Testing
- Acceptance Testing

#### **6. Deployment**

Release working software.

**7. Feedback**

Collect customer feedback.

**8. Maintenance**

Fix bugs and add features.

**Agile Development Cycle**

Requirements



Planning



Sprint



Coding



Testing



Deployment



Feedback



Next Sprint

**Advantages**

- Fast delivery
- Customer involvement
- Continuous improvement
- Lower risk
- High quality

**Disadvantages**

- Frequent customer involvement needed
- Scope may change often
- Requires experienced teams

**C. System Operation**

System Operation is the process of running, monitoring, supporting, securing, and maintaining a deployed system to ensure continuous service.

**Activities****1. Monitoring**

Track system performance.

Example:

- CPU usage
- Memory usage
- Response time

**2. User Support**

Help users resolve issues.

Example:

- Password reset
- Login issues
- Technical support

### **3. Backup**

Regularly back up data to prevent loss.

### **4. Security**

Protect the system.

Activities:

- Authentication
- Authorization
- Encryption
- Firewalls
- Security updates

### **5. Maintenance**

Types:

- Corrective (bug fixes)
- Adaptive (environment changes)
- Perfective (performance improvements)

### **6. Performance Optimization**

Improve speed, scalability, and reliability.

## **Agile System Operation**

Agile often integrates development and operations through **DevOps**.

Practices include:

- Continuous Integration (CI)
- Continuous Delivery/Deployment (CD)
- Automated testing
- Monitoring and logging
- Rapid bug fixes
- Frequent releases

### **Example**

An online shopping website:

- Developers release new features every two weeks.
- Operations team monitors server health.
- Automated tools detect failures.
- Bugs are fixed quickly and deployed in the next release.

### **Advantages**

- High system availability
- Better security

- Faster issue resolution
- Improved user satisfaction
- Continuous improvement